Algorithm 3 LM-TOUR (v5: discs $B(\cdot,\xi)$)

Require: AI nodes with weights w, land graph G, R, ρ, ξ, θ, $L_{\max}$

$A^\star \leftarrow$ AI nodes sorted by w (descending); $\mathcal{K}_1 \leftarrow \emptyset$
for all $u \in A^\star$ **do**
  if $d_G(u, \mathcal{K}_1) \geq R$ **then** $\mathcal{K}_1 \leftarrow \mathcal{K}_1 \cup \{u\}$
clusters $\leftarrow$ GROW$(G, \mathcal{K}_1)$ ▷ balanced Voronoi regions
$\gamma \leftarrow$ Dubins TSP-N through discs $B(u,\xi)$, $u \in \mathcal{K}_1$
$\mathcal{C} \leftarrow \{v \in A^\star \setminus \mathcal{K}_1 : d(v, \text{visited}) > R/2\}$
repeat
  for all $v \in \mathcal{C}$ **do** $\Delta L(v) \leftarrow$ cheapest local re-route of γ into $B(v,\xi)$ passing the window check
  $v^\star \leftarrow \arg\max_v w(v)/\Delta L(v)$
  if $w(v^\star)/\Delta L(v^\star) < \theta$ **or** $|\gamma| + \Delta L(v^\star) > L_{\max}$ **then break**
  insert $v^\star$; remove from $\mathcal{C}$ all nodes within $R/2$ of $v^\star$
until $\mathcal{C} = \emptyset$
return γ, clusters, visited nodes

Window check. Every visited node must be offered a full copy of the query nearby: the stretch of $Q\Delta \approx 2\,\text{km}$ of path around its closest approach must lie within the River reach R_Q of the node ($R_Q \approx 1\,\text{km}$ for one window, from the saturation sweep). Candidates that break it are rejected or the stretch is straightened.

Choosing R. Kept nodes are pairwise at least R apart, so discs of radius $R/2$ around them are disjoint and $|\mathcal{K}_1| \leq 4A/(\pi R^2)$ (plus a boundary term) for an area A. A tour through n points in area A is at most $\sqrt{2nA} + O(\sqrt{A})$ [27], and Dubins turns add $O(n\rho)$, so the base tour grows as $1/R$, while River's delivery distance grows as R. Since $\xi \leq \rho$ and $\xi \leq \sqrt{r^2 - h^2}$, every landmark hears the UAV directly; we use $\xi = 255$, 255, 199 and 99 m at $r = 1000$, 500, 200 and 150 m, and sweep $\xi \in \{100, 175, 250\}$ m.

V. EVALUATION

Setup. ns-3.46 with `lr-wpan` (CSMA/CA, ACK, retransmission); channel and radio as in Section III-B ($\alpha = 3$). Three maps with 8, 12 and 15 clusters (45, 66 and 92 critical cells; capacitated k-medoids, at most 80 cells each). Four A2G regimes: $L_{\text{obs}} = 0$, 9.0, 19.8 and 25.7 dB in (1), calibrated on the ns-3 pass so that $\sqrt{r^2 - h^2}$ is 1015, 498, 199 and 99 m, i.e. $r \approx 1000$, 500, 200 and 150 m. Team 1 flies a speed profile (45° bank, $\pm 2\,\text{m/s}^2$), slowing to 25 m/s or looping where a cluster would not be offered one copy of the query (Eq. (3)); the clue set is fountain-coded and systematic. One ns-3 run per cluster and 100 paired draws of the matching layer per map: every scheme sees the same pass bitmaps, coded receptions and draws. Results are exploratory ($N < 120$).

Schemes. LM-Tour + River-FC (proposed, Section V). *CHORD + River-FC*: the longest flyable chord through each cluster head, bearings and visiting order chosen jointly to minimise the mean time critical cells are left behind. *FC + gather*: team 1 flies a TSP over the critical cells and circles each until its nodes hold enough coded packets; in-cell gathering only. *FC + DTSP-N*: as FC + gather, but nearby critical cells are grouped and served from orbits chosen by a sampled Dubins TSP with neighbourhoods. *Single pass, no cooperation*: CHORD's flight; each critical cell uses only what it decodes itself. *No broadcast (rotary TSP)*: team 2 alone visits every critical cell with the full reference. *Oracle*: team 2 flies straight to the target cell.

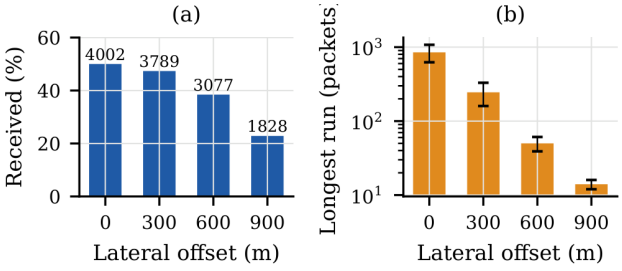


Fig. 4. A2G reception at $r \approx 1\,\text{km}$ (200 passes). (a) Packets received out of 8,001. (b) Longest error-free run. Reception decays slowly with lateral offset; contiguity collapses.

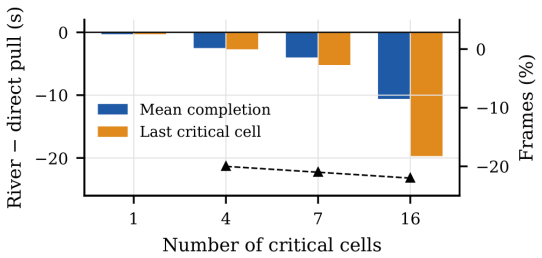


Fig. 5. River minus direct pull on a 91-cell cluster (20 paired passes): completion time (bars) and MAC frames (triangles) versus the number of critical cells.

A. Why cooperation is needed

Even in the favourable regime, a node 900 m from the path receives 1,828 of 8,001 packets but its longest run is 14 packets, against ≈250 for one face image (Fig. 4). Under short range a node hears only its own overhead segment.

B. River versus direct pulling

With 4, 7 and 16 critical cells River uses 20–22 % fewer frames in 20/20 passes and finishes the last critical cell 2.7, 5.2 and 19.7 s earlier (Fig. 5); with one critical cell on the SPINE the two coincide. On cluster C3 (80 cells, four critical cells) River completes the last critical cell in 37.2 s versus 50.3 s (−26 %) with 89.7 k versus 102.2 k frames (−12 %). At mission level the two give the same $T^\star$ within 3 % in 11 of 12 settings; River never stalls, while direct pulling leaves 6 of 92 critical cells incomplete at $r = 150\,\text{m}$.

C. Robustness to the A2G regime

Table I: without cooperation every critical cell completes at $r = 1\,\text{km}$, 66 % at 500 m and almost none at $r \leq 200\,\text{m}$. FC-based schemes complete every critical cell only by circling: the last one completes 2.1–5.2× later than with River-FC as r shrinks, whereas River-FC's completion grows by only 150–350 s, because River sources data from the cells under the path, which hear their own segment at any r.

TABLE I

CRITICAL CELLS COMPLETE AFTER THE PASS (%) AND TIME FROM TAKE-OFF TO THE LAST COMPLETION (S) VERSUS A2G RANGE, $K = 16$ (92 CRITICAL CELLS). [EST]: ESTIMATED.

Scheme	$r = 150$	200	500	1000 m
No cooperation	0 % / –	1 % / –	66 % / –	100 % / –
FC + gather	100 % / 7257	100 % / 6591	100 % / 2784	100 % / 1132
FC + DTSP-N	100 % / 6761	100 % / 5523	100 % / 2529	100 % / 1041
River-FC (CHORD)	100 % / 1395	100 % / 1398	100 % / 1204	100 % / 1052
River-FC + LM-Tour	100 % / 1283	100 % / 1268	100 % / 1082	100 % / 1018

TABLE II

MEAN MISSION TIME $T^\star$ (S), 100 PAIRED DRAWS, K = 8 / 12 / 16 CLUSTERS. [EST]: ESTIMATED, NOT RUN.

Scheme	$r = 1000$ m	500	200	150
LM-Tour + River-FC	647 / 686 / 863	775 / 942 / 1097	822 / 936 / 1143	810 / 942 / 1157
CHORD + River-FC	647 / 686 / 863	784 / 950 / 1107	876 / 975 / 1192	872 / 990 / 1215
FC + gather	660 / 815 / 949	1138 / 1451 / 1801	2285 / 3288 / 3964	2463 / 3460 / 4368
FC + DTSP-N	708 / 805 / 967	1078 / 1388 / 1750	1904 / 2482 / 3469	2190 / 3248 / 4468
Single pass, no cooperation	650 / 687 / 862	1379 / 2030 / 2549	3711 / 5277 / 6298	3860 / 5308 / 6358
No broadcast (rotary TSP)	2926 / 4168 / 5169	2926 / 4168 / 5169	2926 / 4168 / 5169	2926 / 4168 / 5169
Oracle	322 / 402 / 404	322 / 402 / 404	322 / 402 / 404	322 / 402 / 404

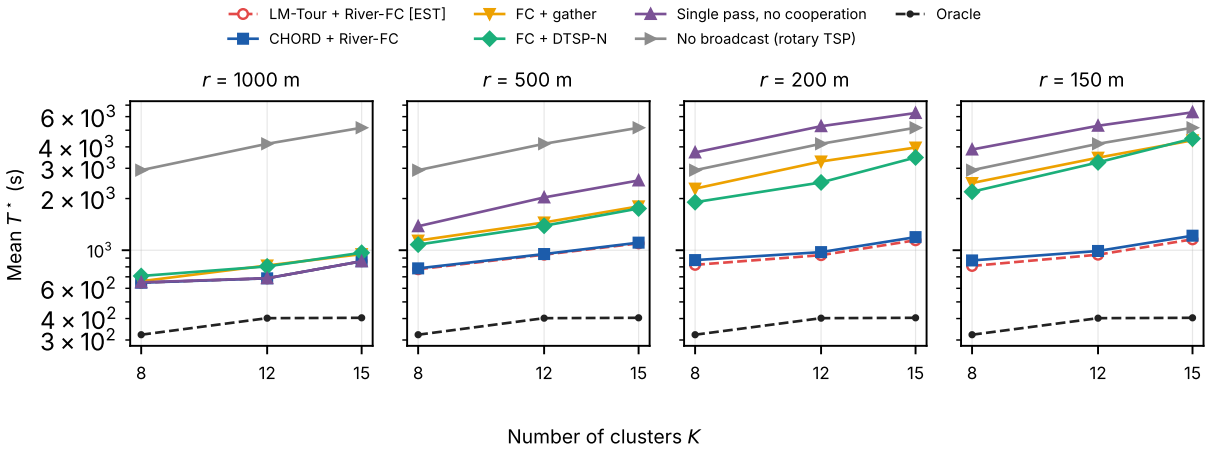


Fig. 6. Mean $T^\star$ versus the number of clusters, per A2G range (log scale). LM-Tour estimated [EST]. Oracle knows the target cell.

D. Mission level

Table II, Fig. 6: the single-pass architecture with River-FC is 3.3–6.0× faster than having no broadcast phase. Against the FC baselines it is 2–15 % faster at $r = 1$ km and 27–72 % faster at $r \leq 500$ m, and the gap widens with K: CHORD's flight stays 782 / 1152 / 1415 s at every r, while FC's grows 2–6×. Without cooperation the same flight is 1.8–5.4× slower at $r \leq 500$ m. [EST] LM-Tour shortens $T^\star$ by a further 1–7 % at $r \leq 500$ m (no change at 1 km), by bringing the path within direct hearing of each landmark.

VI. CONCLUSION

In a single-pass UAV broadcast the flight path is the transmission schedule, so the BS can plan the ground's cooperation before take-off. River turns this into a pre-planned, bidirectional transport that delivers the query to the few AI-capable cameras regardless of A2G range, works with fountain coding, and pulls event frames from legacy cameras over the same routes. Against UAV-served fountain-coded baselines the mission is 27–72 % faster once the A2G range falls to 500 m or below. [EST] LM-Tour adds 1–7 %. [TODO: field A2G measurements.]

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